Joselyn Romero Avila

jromero158486.github.io | Ithaca, NY, US | +1 (530) 220-2458

Education

National University of San Marcos, Lima, Peru — 2020–Jun 2026 (expected)
Bachelor of Biomedical Engineering
Relevant Courses: Biomedical Signals, Artificial Intelligence, Applied Statistics, Medical Imaging

University of Applied Sciences Upper Austria, Hagenberg, Austria — Sep 2024–Feb 2025
Exchange Program – Artificial Intelligence Solutions
Key Courses: Algorithms and Data Structures, Machine Learning, Artificial Intelligence

Memorial University of Newfoundland and Labrador, St. John's, Canada — Jan–Apr 2024
Exchange Program – Electrical and Computer Engineering
Key Courses: Biorobotics, Computer Science

Research Experience

Hybrid Body Lab – Cornell University (Ithaca, USA) — Jul 2025–Present
Health AI Research Intern — Advised by: Dr. Cindy Hsin-Liu Kao
Worked on *Multi-Agent LLMs for Physiological Signal Interpretation*.

- Developed a multi-agent LLM framework to interpret multimodal biosignals (PPG, IMU, audio), improving adaptability to diverse stress and environmental conditions.
- Built real-time sensor pipelines (microcontrollers, preprocessing, cloud inference) optimized for latency and fidelity, enabling closed-loop wearable feedback.
- Prototyped emotion-aware interfaces and ran usability tests, informing co-creation workshops and participatory design.

IEEE SigMA Program – Cambridge University (Remote, UK) — Jan 2025–Present
Audio Processing Research Intern — Advised by: Dr. Jing Han
Worked on *Deep Learning for Fetal Phonocardiography*.

- Built a hybrid pipeline for respiratory sound analysis combining signal processing (segmentation, spectral features) with LLM-based querying and verification.
- Applied Librosa, SciPy, and ML frameworks (scikit-learn, PyTorch, TensorFlow) to classify multi-event patterns in the ICBHI dataset.
- Designed verification mechanisms to reduce LLM hallucinations and drafted the first manuscript on clinical decision support.

Interactive Organisms Lab – University of California, Davis (USA) — Feb–May 2025
Visiting Undergraduate Research Assistant — Advised by: Dr. Katia Vega
Worked on *Biosensing Wearables for Cortisol Monitoring*.

- Prototyped cortisol-monitoring wearables integrating microfluidic sensors and embedded electronics.
- Applied probabilistic ML to classify stress from multimodal signals with improved accuracy and uncertainty estimation.
- Conducted usability testing to guide HCI design and real-world deployment.

Treetop Medical – AIST Lab (Austria) — Sep 2024–Feb 2025
Natural Language Processing Research Intern — Advised by: Dr.Gerald Adam Zwettler
Worked on *Retrieval-Augmented Generation for Clinical Document Understanding*.

- Developed a RAG pipeline for multimodal clinical documents using advanced chunking and metadata-aware indexing.
- Integrated multilingual LLMs (English/German) for entity extraction and summarization with spaCy and HuggingFace.
- Collaborated with clinicians to refine usability and interpretability; ensured reproducibility via GitHub workflows.

Quantum Valley Navigation Project – University of Waterloo (Canada) — May–Aug 2024
Undergraduate Research Assistant — Advised by: Dr. David Cory
Worked on *Resonator Design and Bayesian Modeling for Quantum Biosensors*.

- Designed and modeled resonator architectures for quantum biosensors, applying Bayesian methods to quantify uncertainties.
- Built high-fidelity simulations in Python, MATLAB, and COMSOL, optimizing calibration and reducing computation time.
- Developed and validated hardware prototypes, aligning experimental results with theoretical predictions.

Google Explore LATAM Research Program – Monterrey, Mexico (Remote) Feb–Aug 2024
Computer Vision Intern Advised by: Dr. Gilberto Ochoa-Ruiz
Worked on *Multi-Instance Learning for Kidney Stone Classification.*

- Developed a MIL framework in TensorFlow/Keras for kidney stone classification with weakly labeled datasets.
- Integrated attention-based pooling to improve accuracy and interpretability by highlighting key image regions.
- Validated robustness under class imbalance via augmentation, cross-validation, and ablation against CNN baselines.

Recovery and Performance Lab – Memorial University (Canada) Dec 2023–Apr 2024
Undergraduate Research Intern Advised by: Dr. Michelle Ploughman
Worked on *Kinematic Biomarkers of Multiple Sclerosis Using Robotic Assessment.*

- Collected high-resolution kinematic data from MS patients using the Kinarm robotic platform.
- Applied ML algorithms (RF, SVM, NN) to identify movement patterns and link robotic metrics with clinical tests.
- Co-authored a peer-reviewed publication validating robotic biomarkers of upper-extremity dysfunction in MS.

Publications

N. W. Bray, S. Z. Raza, **J. Romero Avila**, C. J. Newell, and M. Ploughman. *Robotic Rigor: Validity of the Kinarm End-Point Robot Visually Guided Reaching Test in Multiple Sclerosis. Multiple Sclerosis Journal*, 2024.

Under Review

J. Romero. *RespSound-QA: A Hybrid Pipeline for Respiratory Sound Analysis with Natural Language Querying.* Submitted to IEEE International Conference on Acoustics, Speech, and Signal Processing (ICASSP 2026), under review, 2025.

Presentations

J. Romero, et al. *FoldLock: An Origami-Inspired Bi-Stable Clip for Modular Soft Interfaces.* 2025 IEEE International Conference on Robotics & Automation (ICRA) — Multi-Stable and Origami-based Soft Robots Workshop, 2025.

J. Romero, et al. *Adaptive Terrain Classification for Robotic Prostheses.* Building Responsive Body-Machine Interfaces With Biosignals and Robotic Exoskeletons Workshop — IEEE RAS EMBS 10th International Conference on Biomedical Robotics and Biomechatronics (BioRob 2024), 2024.

Selected Projects

VisionEx: Retinopathy Diagnosis via Mobile Computer Vision HackDavis 2025 (Honorable Mention)
Worked on *Lightweight Mobile AI for Diabetic Retinopathy Detection.*

- Built lightweight CNN for real-time retinal image classification on mobile.
- Designed Figma-based UI for low-resource usability.
- Added natural language explainability via OpenAI API.

Salus: Conversational Platform for Preventive Health and Sleep Apnea MIT Hacking Medicine 2025 (1st Place, Google Health AI Track)
Worked on *LLM-based Conversational Health Assistant.*

- Developed LLM-based assistant with retrieval-augmented responses for preventive health.
- Integrated Google Health AI's HeAR model for sleep apnea detection from wearable data.
- Delivered functional demo and prototype showcased in competition finals.

Awards

2025 Cleveland NeuroDesign Workshop, *Selected as 1 of 30 innovators worldwide from 2000+ applicants representing Latin America for excellence in bio entrepreneurship in neurotechnology.*
2025 REPU Computer Science – UC Davis, USA, *Selected as 1 of the few Peruvian undergraduates nationwide chosen to pursue an international research internship in Computer Science abroad.*
2025 IEEE RAS SPIRSE – ICRA Atlanta, USA, *Awarded SPIRSE grant representing Latin America, contributing to biomedical robotics standards and advancing inclusive global robotics.*
2025 MIT Hacking Medicine – Google Health AI, 1st Place, *Won Google Health AI Prize at MIT Media Lab with clinicians, engineersby creating a system for heart disease detection.*
2024 Ernst Mach Grant – Austria, *Awarded a merit-based full scholarship to study one semester the Artificial Solutions Program at FH Upper Austria.*
2024 IEEE RAS WIE IDEA Travel Grant – BioRob, Germany, *Selected as 1 of 10 global awardees to attend BioRob 2024, advancing inclusion and leadership for women in biomedical robotics.*
2024 Emerging Scholar Advancement Award (ESAA) – Neuronovember, *Selected as 1 of 25 global or outstanding potential in neuroscience and neurotechnology, receiving full support to present and engage in international mentorship.*
2024 5-Minute Thesis, 2nd Place – Brainwaves, Canada, *Recognized for research communication skills as the only undergraduate finalist, competing against master's and PhD students.*
2023 Emerging Leaders in the Americas Program (ELAP), *Awarded a fully funded academic exchange in Canada for research collaboration and international leadership in science.*
2021 PRONABEC Permanencia Scholarship – Peru, *Awarded a full-ride scholarship (tuition, housing, services, and transportation) as part of the top 1% of students, recognizing sustained academic excellence.*

Leadership and Activities

IEEE Signal Processing Society (SPS) UNMSM, *Co-Founder & Vice Chair* 2024–Present
IEEE Brain Peru, *Co-Founder & Marketing Director* 2023–Present
IEEE Engineering in Medicine and Biology Society (EMBS) UNMSM, *Vice Chair* 2023
IEEE EMBS UNMSM, *Director of Logistics & Treasurer* 2022
IEEE Women in Engineering (WIE) UNMSM, *Member* 2022–Present
Biomedical Engineering Student Council (UNMSM), *Contributor* 2022
IEEE EMBS Region 9, *Volunteer* 2023–Present
AI in Health Journal Club (UPCH), *Participant* 2022

Technical Skills

Programming, *Python, SQL, Bash, C++, R (basic), MATLAB, JavaScript (basic)*
Machine Learning & AI, *TensorFlow, PyTorch, Keras, Scikit-learn, Hugging Face Transformers, Transfer Learning, MIL, RAG, Prompt Engineering, SHAP, Grad-CAM*
Data Processing & Analysis, *Pandas, NumPy, SciPy, OpenCV, SpaCy, Librosa, unstructured data pipelines, signal processing (ECG, EEG, PPG, GSR), statistical analysis (t-test, ANOVA), visualization*
Systems & Tools, *Jupyter, Git/GitHub, Linux Terminal, PyCharm, VS Code, AWS (basic), Google Cloud (basic), MongoDB, SQL Server, REST APIs, Twilio, LangChain, Zapier*
Hardware & Design, *COMSOL Multiphysics, SolidWorks, PCB design, Arduino/ESP32, Unity (AR/VR), 3D printing, basic FPGA programming*
Languages, *Spanish (native), English (fluent), German (basic)*